

# Allocation Score Manuscript

Draft 1

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## Abstract

In the development and assessment of infectious disease forecasts, modelers stand to benefit from evaluation measures that express a forecast’s success in scenarios of public health relevance. The integrated allocation score (IAS) is a scoring rule that measures the units of unaddressed resource need in an abstract scenario where a forecast directly informs allocation decisions. COVID-19 weekly incident case forecasts for 57 counties in California were compiled in quantile format from the COVID-19 Forecast Hub and assessed by IAS over 13 weeks of the COVID-19 omicron wave at the 1- and 3-week horizons, using resource constraints informed by ground truth case counts. At each target date, an epidemiologically-naïve resource allocation based on county populations was scored as a control IAS. Model rankings by IAS were found to align broadly with rankings by mean weighted interval score (MWIS), a commonly used scoring rule for probabilistic forecasts. Comparison of model IAS values to the control, however, found that forecast-informed allocations frequently failed to outscore the population-based allocation, particularly at the highest-burden weeks in the epidemic. The incorporation of IAS into the evaluation and optimization of forecasts may therefore be of future benefit to forecasters, particularly when developing predictions for scenarios in which forecasts are used by officials to inform resource distribution decisions across multiple locations.

## Introduction

### Forecast Use in Decision-Making and Need for Use-Aware Evaluation Metrics

In recent years, infectious disease forecasting has been used increasingly in public health surveillance and outbreak response. Accurate, timely forecasts of infectious disease indicators provide state, tribal, local, and territorial (STLT) public health practitioners with advance notice of local and national trends, enabling for rapid response to disease burden spikes, both through pharmaceutical and nonpharmaceutical interventions and through public information-sharing. Well-performing forecasts have the potential to contribute substantially to existing arsenals of public health tools, increasing the effectiveness of agencies’ health-promotion efforts. Most notably, infectious disease forecasts came into widespread usage during the COVID-19 pandemic, through projects such as the COVID-19 Forecast Hub (“Home - COVID 19 Forecast Hub,” n.d.). Additional large scale examples of forecast adoption include CDC FluSight (CDC 2026) and the Dengue Forecasting Project (Johansson et al. 2019), in which forecasts submitted by modeling groups were compiled, assessed, and used to inform both local and national public health policy. Modeling groups themselves recommend use of their forecasts for a range of objectives, including public communications, advance notice of spikes for hospitals and health departments, and to aid decisions around the allocation of

medical resources and staffing.(Lutz et al. 2019) Indeed, a recent review of COVID-19 forecast usage found that, during the COVID-19 pandemic, jurisdictions applied these forecasts to uses such as understanding transmission dynamics, directing planning, and guiding resource management and allocation. (Rao et al. 2025)

However, as jurisdictions increasingly adopt these tools, a number of issues remain, and of these, the potential for misalignment between the priorities of infectious disease modelers and STLT practitioners is of particular relevance.[CITE] Misaligned goals between these two groups risks the development of models that fail to answer relevant public health questions, or that prove unsuitable for use in high-stakes policy decision-making due to incorrectly defined forecasting targets, unreliable predictions, or lack of interpretability.(Metcalf, Edmunds, and Lessler 2015) For forecast implementation to be productive, it is therefore necessary that models be created, evaluated, and adopted with both the public health need and the priorities of the acting public health agency in mind.

Forecast evaluation metrics that align closely with public health goals are a vital part of this alignment. In order to ensure that models perform well in practice, forecast scoring metrics are used to evaluate and rank forecasts, based on comparisons between predicted outcomes and ground truth results. A range of methods exist to conduct these evaluations, each with differing assumptions and objects of measurement.(Bjerregård, Møller, and Madsen 2021) Any definition of a “best” forecast is dependent upon which qualities the evaluation metric values most – this may include the closeness of point estimates to the ground truth, the width of prediction intervals, or the correct relative ranking of predicted values across a group of targets.[CITE CITE CITE] For this reason, forecast evaluation should include metrics that approximate user priorities, so that the best-scoring models in the testing phase are indeed those the most suited to the forecasting use of interest.

In this case study, we present an example of the application of the integrated allocation score (IAS), an evaluation metric that scores infectious disease forecasts based on effective resource allocation across multiple forecast targets, in the context of the COVID-19 omicron wave in California. We hope that this study can serve as an example of IAS’s usage in a highly uncertain public health context, and reveal its qualities through contrast with other scoring methods.

## **Allocation Score and Integrated Allocation Score**

High accuracy and precision in forecasts of individual targets (ex. a single county) may not necessarily translate to optimal decision-making in dynamic scenarios with multiple allocation targets (ex. an entire state). The distribution of resources under constraint is a high priority for public health policy-makers and decision-makers in epidemic contexts, such as in scenarios of personal protective equipment (PPE) distribution [CITE]. The allocation score (AS), introduced by Gerding et al.(Gerding et al. 2024), scores forecasts on their success in allocating limited resources across multiple locations.

For a given target date, this method interprets a forecast’s predictions as location-specific estimates of resource need and uses these predictions to guide the allocation of a fixed quantity of resource. The forecast is then scored by comparing this forecast-informed distribution scheme to an “oracle distribution” produced from the the ground truth (ie. true resource need): for both distributions, the total resource underallocation across all locations is tallied, and the difference between the resource underallocation of the oracle distribution (considered to be unavoidable) and that of the forecast-informed distribution is the allocation score. Lower scores reflect less avoidable underallocation of resources by the forecast and therefore better predictions. This negative orientation aligns with other commonly used forecast scoring metrics, such as WIS.(Bracher et al. 2021)

To address uncertainty with regards to resource constraints, the integrated allocation score (IAS) is preferred over AS alone. IAS is determined by calculating AS across a series of possible resource constraints and

producing an average, which has the option of being weighted based on the perceived probabilities of the possible resource constraints, or unweighted using a uniform distribution.

In utilizing IAS, it is important to be aware of a few important attributes and their impacts on the score. First, it should be noted that the IAS method measures underallocation, not overallocation, in the distribution of resource to the locations, meaning that it does not penalize resource waste. Second, the measurement of total underallocation as sum of all targets means that locations with larger populations and/or resource needs will contribute more to the score than smaller ones, an assumption that is valid in field situations, but potentially disadvantages forecasts with strong performance in smaller locations. Third, AS does not directly account for the predictive distribution of the forecasts being evaluated (see [Methods](#)), and so sharper predictions are not rewarded. Finally, in this scheme, the correct prediction of absolute values is secondary to the capture of relative ranking, meaning that a forecast which predicts every location as having exactly double the ground truth need would score better on the AS than one where all but a one location was predicted perfectly.

### Example: Allocation Score of the 1-week COVIDhub-baseline forecast for January 1st, 2022

As a illustrative example, we will visualize the calculation of an allocation score for the COVIDhub-baseline forecast for the target date of January 1st, 2022 at the 1 week horizon, using country-level COVID-19 case forecasts for the State of California, with the exception of Los Angeles County. For the purpose of scoring, predictions of COVID-19 incident cases are used here as proxies for resource need and the resource is kept abstract, but assumed to be apportionable. For this example, we set a resource constraint equal to 90% of the total incident cases for the date of interest: for the week ending 01/01/2022, a total of 131,492 cases were reported, and so the resource constraint for this target date is set at 118,343 units of resource, 90% of this. These resources are distributed based on the county-level incident case predictions of the COVIDhub-baseline forecast for this target date and horizon. (See [Methods](#) for more details on the distribution of resources from a forecast in quantile format.) In Figure 1, we see a breakdown of this resource allocation, in which red shading indicates underallocation to a location, blue shading indicates overallocation, and purple shading indicates addressed need.

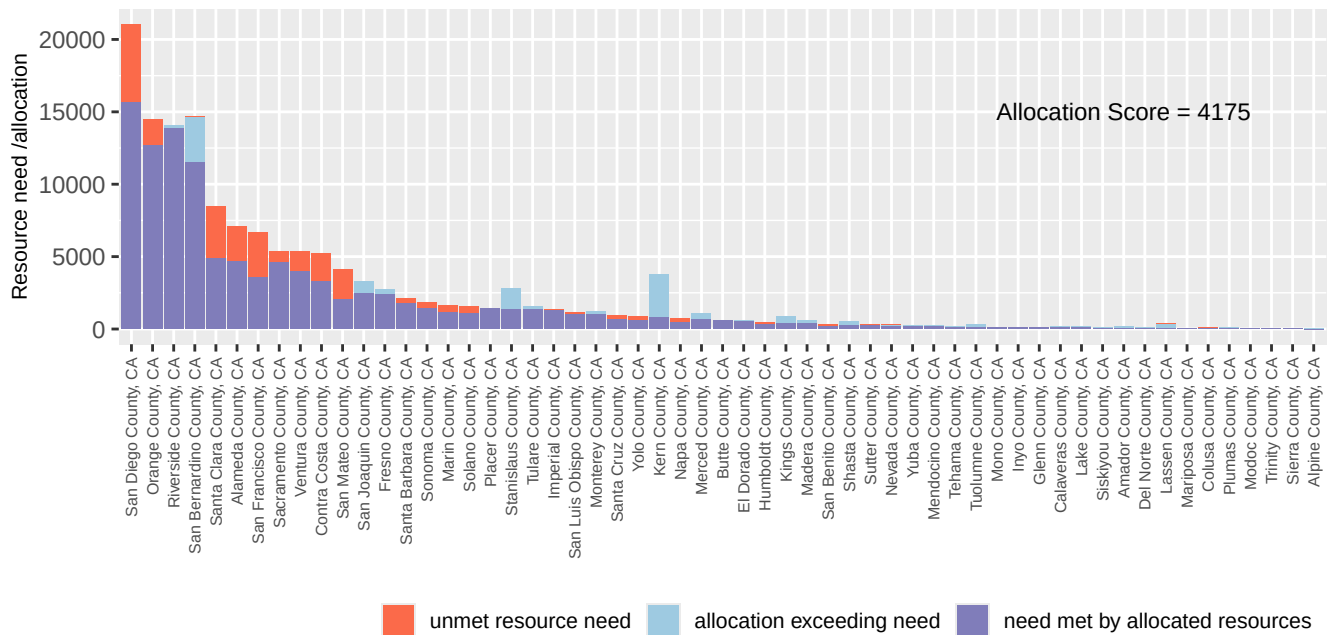


Figure 1: Under- and Overallocation of Resources Based on COVIDhub-baseline Forecast for 01/01/2022. Resource constraint of 90% of true need, forecasting at the 1 week horizon

Overall underallocation is calculated by summing this red-shaded underallocation across all counties (17,324 units of resource). On a similar graph of the oracle distribution (not shown), the red shaded areas will represent unavoidable underallocation, as the oracle distribution is assumed to be ideal. Given a resource constraint of 90%, it follows that an oracle scheme will inevitably underallocate resources by 10%, and so unavoidable underallocation is known to be 10% of true need (13,149 units). Allocation score is then calculated as the difference between these two quantities (total underallocation minus unavoidable underallocation; 4,175) to estimate avoidable resource underallocation due to the forecast predictions.

Notably, overallocation (shown in blue) is not directly counted by AS, but due to resources being finite, a forecast with a large amount of overallocation will inevitably have a worse allocation score, due to incorrectly distributed resources.

In the visualization above, marked overallocation of resources to San Bernardino, Stanislaus, and Kern Counties can be identified, as well as underallocation to San Diego County and the counties from Santa Clara County to San Mateo County on the graph. The use of the allocation score framework and its associated visualizations has the added benefit of facilitating such qualitative assessments and can therefore assist in identifying targets for which a forecast's predictions are weakest. Review of these results can help to identify data artifacts or other threats to forecast accuracy. **[CA PEOPLE KNOW THE SOURCE OF THESE OVERPREDICTIONS? COULD POINT TO AS ILLUSTRATIVE]**

In practice, decision-makers may lack a definitive estimate of true resource availability. Furthermore, the calculation of AS from a single resource constraint may fail to capture resource distribution dynamics at higher or lower constraints. For this reason, the integrated allocation score is produced as an average of a range of possible resource constraints, which can be weighted based on perceived probability of each constraint. For this example, an unweighted mean (uniform weighting) of a 41 allocation scores from 70-110% of true need (92,044-144,642 units of resource) were calculated to produce an IAS of 12,284. As with AS, lower scores of IAS are indicative of better performing forecasts, and so this score may be compared to those of other forecasts for the same target date to assess model performance.

## Methods

In this case study, forecasts of county-level incident COVID-19 cases during the omicron wave were evaluated for the state of California using IAS and MWIS.

## Data Sources

Forecast data were sourced from the COVID-19 Forecast Hub, a central repository of forecasts and predictions submitted by international research groups. ("Data - COVID 19 Forecast Hub," n.d.) Submissions are compiled in a standardized format for defined targets, locations, and time horizons, allowing for visualization, scoring, and analysis of forecasts.

Forecasts from the Hub's Zoltar repository **[CITE]** were extracted for a total of 13 target dates covering the duration of the omicron wave (12/18/2021 to 03/12/2022; MMWR weeks 2021-50 to 2022-12), at both the 1 and 3 week horizons. Target of interest was the county-level weekly incident COVID-19 case counts for all California counties with the exception of Los Angeles County (n=57), which was excluded due to the existence of significant resource lines from the federal government **[CITE]**. Since most other counties in the state receive a majority of their resources from the State of California, Los Angeles County's allocation framework was deemed significantly different from the other counties as to justify its exclusion from the analysis. Through dropping this county, resource allocation could be conceived as coming from one centralized decision-maker, and COVID-19 case count could be used as a direct proxy for resource need.

Ground truth counts of county-level incident cases were retrieved via the `load_truth()` function in the `covidHubUtils` package, using the the JHU repository.<sup>[CITE]</sup> For the calculation of population-based resource allocations, country-level population estimates for January 1st, 2021 were sourced from the State of California Department of Finance. (“E-4 Population Estimates for Cities, Counties, and the State, 2021-2026 with 2020 Census Benchmark | California Department of Finance,” n.d.).

## Model Selection

Forecast submissions were deemed eligible for analysis if they included predictions for all 57 counties of interest on a given target date and horizon. Additionally, redundant Forecasting Hub ensembles were dropped (COVIDhub-4\_week\_ensemble was retained) as well as redundant models from Columbia University.

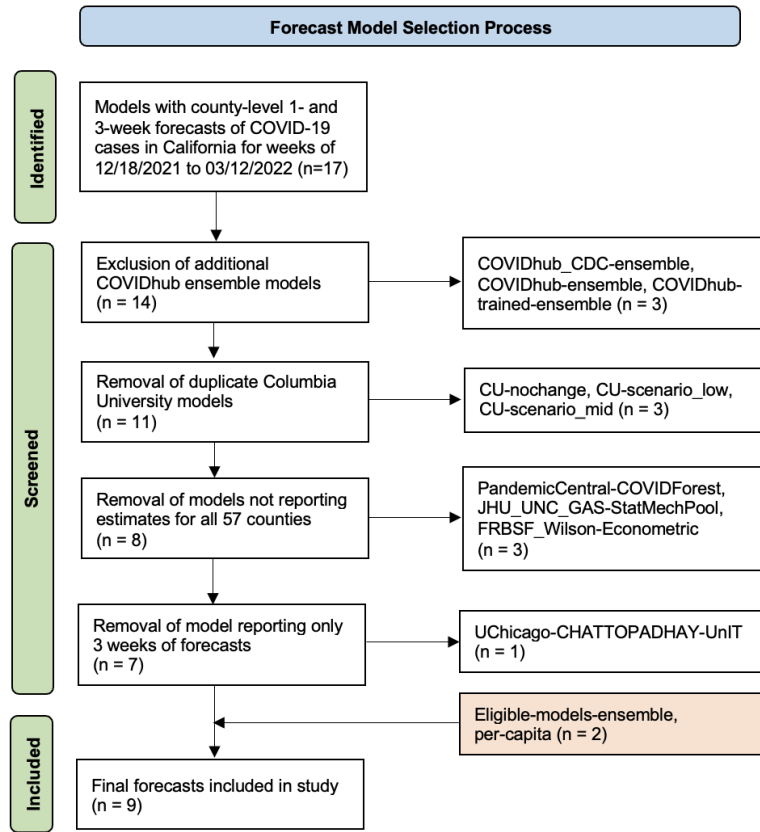


Figure 2: Forecast Model Selection Process

In addition to eligible individual forecast submissions, an ensemble forecast, named the “eligible-models-ensemble,” was generated as an unweighted average of all models with eligible submissions for each respective target end date and horizon, excepting the Hub’s existing ensemble forecast (COVIDhub-4\_week\_ensemble). Quantile estimates for the eligible-models-ensemble were produced as an unweighted average of qualifying submissions for that quantile - for example, to produce a 0.90 ensemble estimate for a given county, horizon, and target date, the mean was calculated from the 0.90 estimates from all eligible models submitting for that county, horizon, and date.

Finally, a “per-capita” allocation scheme, in which the units of resource were allocated proportionally to estimated county population, ignoring epidemiological data, was evaluated as a control.

## Statistical Analysis

### Resource Distribution from Probabilistic Forecasts

In order to score the forecasts, optimized resource allocation schemes had to be produced from predictions reported in a quantile format. The method of resource allocation included in the `alloscore` package's `alloscore()` function was utilized for this process.(Gerding et al. 2024)

In this method, first, the set of quantiles submitted to the Hub is used to interpolate a cumulative distribution function (CDF) of expected case counts for each forecast submission. Next, using the calculated CDFs of the locations of interest and a set resource constraint, an iterative process is used to solve for a forecast-informed allocation of resource across the locations. At a given probability level, the predicted case counts from the CDFs are summed over all locations to determine the total predicted resource need at the probability level. The algorithm then iterates, increasing or decreasing the probability level across CDFs in parallel to solve for a probability level that approximates the total resource constraint. The resulting solution, expressed as a set of predicted case counts (i.e. predicted resource needs) across locations at the optimized probability level, are considered the forecast's allocation scheme for the target date and horizon. These allocation schemes are then scored using the allocation score procedure.

### Integrated Allocation Score

For each respective target date and horizon, integrated allocation scores for forecasts were calculated using resource constraints ranging from 70% to 110% of ground truth case count (true need) at the respective target date (one allocation score for each percent between 70% and 110%, centered at 90% of true need,  $n = 41$ ). A uniform-weighted integrated allocation score was used for most analyses, calculated as a mean of the 41 allocation scores. To evaluate the impact of probability weighting on IAS, uniform-weighted scores were compared to a normal-weighted score, with a distribution centered on the mean resource constraint (90% of true need) and a standard deviation calculated from the set of 41 resource levels.

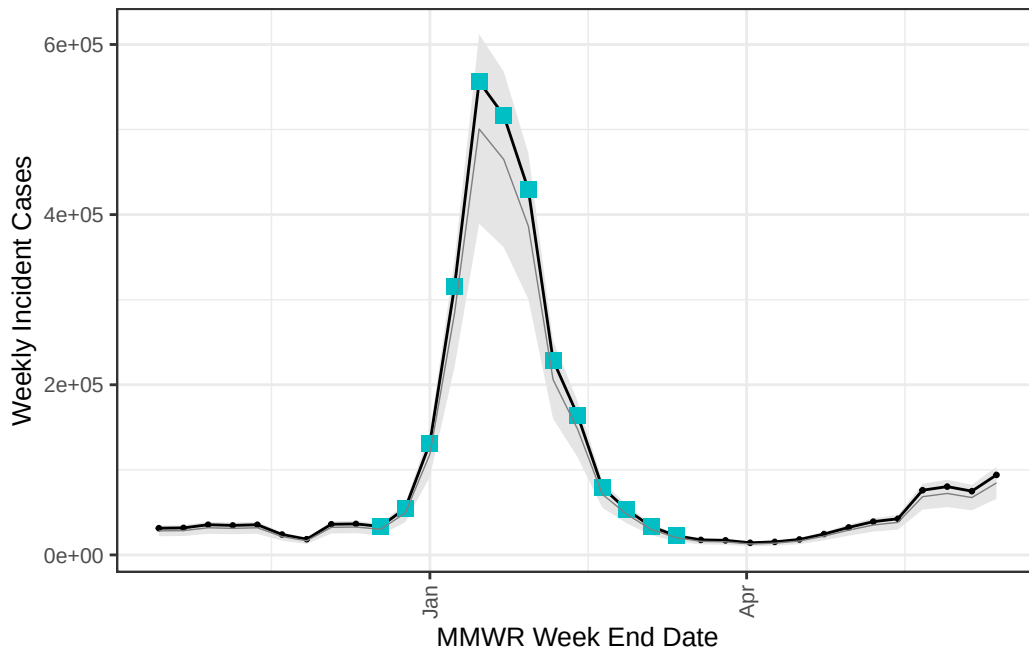


Figure 3: Omicron Wave in California: Allocation Score Resource Constraints (90% of true need, 70%-110% of true need)

## Pairwise Comparisons

In addition to uniform-weighted IAS and normal-weighted IAS, forecasts were evaluated via weighted interval score (WIS) for each target date, county, and horizon. (Bracher et al. 2021) Mean WIS (MWIS) was calculated as the unweighted average WIS score across all 57 counties for a given target date and horizon. Kendall's  $\tau$  was used to evaluate agreement in relative rankings of models between scoring metrics at the two horizons.

## Technical

All analyses were run in R version 4.6.1 and **targets** package version 1.11.3 (Landau 2021). AS and IAS were calculated using the **alloscore** package (Gerding 2026) and WIS scores using the **scoringutils** package (Bosse et al., n.d.).

## Results

### Model Availability

At the 1 and 3 week forecast horizons, a total of 9 forecasts/schemes were evaluated, including the per-capita and eligible-models-ensemble schemes. Figure 4 outlines eligible forecast submissions (defined as submissions providing the required quantiles for all 57 counties of interest) by target date. Following model selection, all eligible submissions were used in the analysis. With the exception of the COVIDhub-4\_week\_ensemble, all eligible models for a given target and horizon were incorporated into the eligible-models-ensemble.

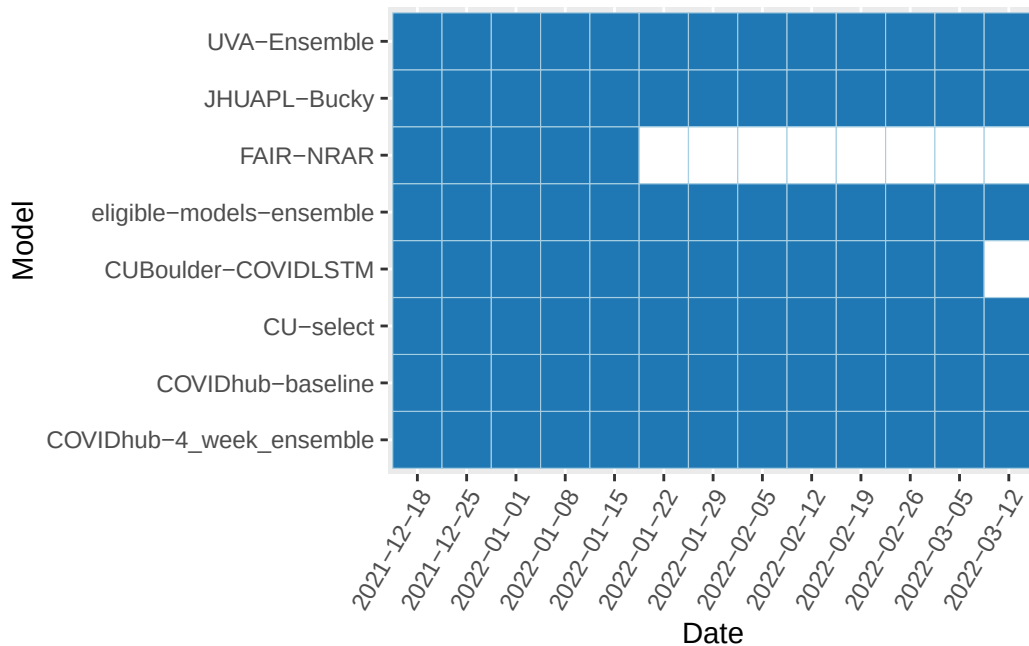


Figure 4: Complete Forecast Submissions by Model and Date

## Integrated Allocation Score

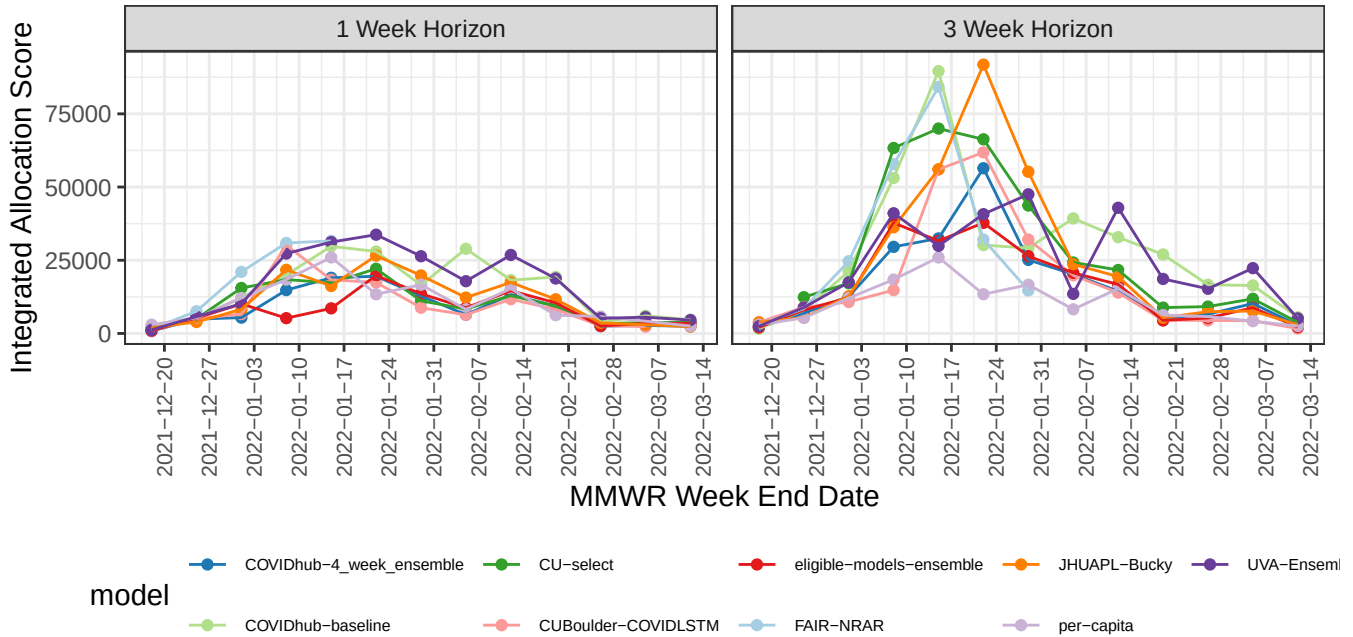


Figure 5: Integrated Allocation Scores Over Time, by Horizon - Uniform-Weighted Average of 70% - 110% Resource Constraints

Uniform-weighted IAS results were found to be noisy (Figure 5), but largely scaled with the epidemic curve. **[NOISY IS WRONG WORD - WHAT WORD TO USE?]** At the 1-week horizon, the IAS values were typically found to be lower than those at the 3-week horizon and models scored better than the per-capita control more frequently (53.7% at the 1-week horizon, 18.6% at the 3-week horizon). At the 1-week horizon, 5 of the 8 submitting models (COVIDhub-4\_week\_ensemble, CUBoulder-COVIDLSTM, eligible-models-ensemble, CU-select, JHUAPL-Bucky) outsourced the per-capita allocation in more than half of their predictions. In contrast, at the 3-week horizon, no model scored better than the per-capita distribution in more than half of its predictions. Of the models, only two models scored better than the per-capita allocation in more than half of their predictions over the two horizons: CUBoulder-COVIDLSTM (60%) and COVIDhub-4\_week\_ensemble (53.8%) .

Of the forecast target dates, the December 18th target had the highest proportion of predictions (13 of the 15 predictions, 86.7%) scoring better via IAS than the per-capita allocation, followed by February 26th (9 of the 14 predictions, 64.3%). In contrast, forecast predictions for all other target dates scored better than the per-capita allocation less than half of the time. Disparities between IAS values of the per-capita allocation and the other models were particularly dramatic at the 3-week horizon during the peak weeks of the epidemic, when the IAS values for many models were 3 or more times that of the per-capita distribution.



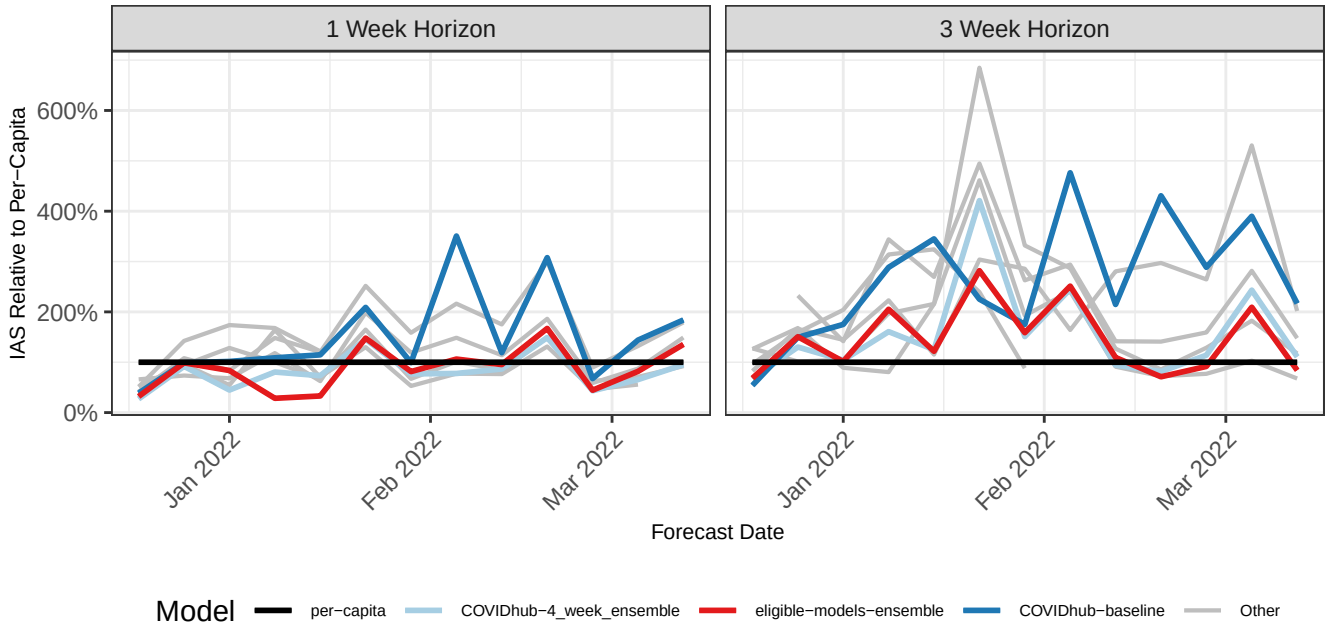


Figure 6: Uniform-Weighted Integrated Allocation Scores Relative to Per-Capita Allocation, by Target Date

Figure 7 displays the IAS values as a percentage of the IAS of the control per-capita distribution, with select models highlighted (COVIDhub-4\_week\_ensemble, eligible-models-ensemble, and COVIDhub-baseline). In this visualization, lines under the bolded black line (representing 100% of the per-capita IAS) indicate better allocation performance. Both ensemble models were found to frequently score lower than the per-capita distribution at the 1 week horizon, but not at the 3-week horizon. Scores were notably higher than the per-capita score on January 22nd (the peak of the epidemic) at the 3-week horizon, with all forecast IAS value at least double that of the per-capita distribution. The COVIDhub-baseline model was found to be outscored by the per-capita allocation and both ensembles for nearly all targets, but disparities were particularly dramatic at the 3 week horizon.

## Pairwise Comparisons

Pairwise comparisons between the different scoring metrics found relative score rankings via 90% AS, IAS, and MWIS to be broadly in alignment via Kendall’s  $\tau$  [Table 1]. Furthermore, rankings in scores at the 1-week horizon were found to align with rankings of scores at the 3-week horizon for all three metrics (90% AS:  $\tau = 0.802$ ; IAS:  $\tau = 0.645$ ; MWIS:  $\tau = 0.711$ ).

Table 1: Kendall’s  $\tau$  values for pairwise tests

| Comparison | 90% AS vs. IAS | 90% AS vs. MWIS | IAS vs. MWIS |
|------------|----------------|-----------------|--------------|
| 1 week     | 0.5649013      | 0.5861142       | 0.7083987    |
| 3 week     | 0.6266889      | 0.6254296       | 0.7014605    |

## Uniform- vs. Normal-Weighting in IAS Calculation

Kendall’s  $\tau$  was used to evaluate agreement between model score rankings using IAS calculated via uniform weighting (mean) versus a normal distribution. Weights were centered on the mean resource constraint (90% of true need) and standard deviation was calculated as standard deviation of the set of resource

constraint values used. IAS values between the two methods were found to be highly concordant in their model rankings, with  $\tau$  values of 0.988 at the 1 week horizon, and 0.993 at the 3 week horizon.

## Discussion

Compared with MWIS and AS at a 90% resource constraint, IAS values were found to be strongly positively associated with the other scoring measures. In particular, comparison of relative rankings via Kendall’s  $\tau$  found particularly close agreement between scores by uniform-weighted IAS and MWIS. Additionally, a comparison between IAS calculated via uniform- versus normal-weighting found the two metrics to rank forecasts essentially the same. **[TOO CASUAL?]** With the exception of CU-Select (which was among the best by MWIS at both horizons but underperformed other models in IAS), forecast performance in IAS and MWIS were found to agree broadly, in spite of the two methods using very different evaluation criteria in their measures. Taken together, this broad alignment in rankings indicates agreement between methods and weighting strategies on which models performed “better” for the given targets. Of the forecasts evaluated, the ensemble models (COVIDhub-4\_week\_ensemble and eligible-models-ensemble) and CUBoulder-COVIDLSTM were among the highest ranking on average between both IAS and MWIS. This finding is in general alignment with findings with other forecasting metrics, as ensemble models are generally considered to outperform their constituent models at forecasting tasks **[CITE]**

However, when compared to a control “per-capita” resource distribution based solely on population and agnostic of epidemiological trends, the forecasts largely failed to outperform the control distribution on IAS. The worst performance, when visualized as IAS relative to the IAS of the per-capita distribution, was found to occur with target dates at the peak of epidemic, when public health decisions can be considered most critical due to high resource burdens and morbidity. During the periods prior to the epidemic and during the decrease, relative IAS results were mixed. This frequent failure of the models to beat the IAS of the per-capita distribution at critical points suggests that forecast-informed resource distribution in the counties of interest may have been of little value-added during the COVID-19 omicron wave.

Out of the models assessed, the two ensemble models assessed had some of the lowest scores and outscored the per-capita distribution with the highest frequency. While these models still underperformed during peak times in the epidemic, the forecasts demonstrated possible utility in allocation during the upswing and final weeks of the epidemic, particularly when forecasting at the 1 week horizon. IAS values at the 1-week horizon were found to be the lower than those at the 3-week horizon and more likely to outscore the per-capita distribution, implying that resource distribution decisions made with shorter-term forecasts more accurately addressed ground-truth need than those further in advance.

## Conclusions

In this analysis, we demonstrate the use the integrated allocation score to score forecasts of county-level incident COVID-19 case data during California’s omicron wave. Relative rankings of models by IAS were found to largely agree with those by MWIS, a frequently-used scoring rule. Additionally, comparison of mean-weighted IAS to scores by normal-weighting determined that weighting method made little difference in relative IAS rankings of models. The evaluation of the scores relative to a control allocation scheme agnostic of epidemiological trends (per-capita) allowed for measurement of the value-added in using the forecasts in resource distribution decisions, compared to solely population-based decision-making. Forecasts frequently failed to beat this control when evaluated by IAS, particularly at the height of the epidemic. From this, we infer that, compared to population-based allocation, the county-level forecasts would not have improved allocation decisions during this critical time period in the public health response. During the

upswing of the epidemic and last weeks of the wave, however, these relative scores were mixed, suggesting possible value-added by some models.

This analysis contains several limitations. First and foremost, scoring by the integrated allocation score, while it attempts to ground forecasts in a decision-making context, still describes only abstract resources and distribution scenarios. Constraints used in this analysis were not intended to pertain to any specific apportionable resource, such as medicine or staffing, but rather were abstract units determined post-hoc from ground truth case counts and presented on the same unit scale. The translation of abstract resource units into real resources runs the risk of introducing feedback loops into the scoring method thought experiment. For example, a forecast-informed distribution of PPE could impact disease transmission in the subsequent weeks and quickly render earlier forecasts invalid. Because of this, the scores cannot be translated easily into waste in units of funding, person-hours, or other resources. Further research would be necessary in order to bridge the gap between theoretical IAS and units of real world resources. However, qualitative assessments via an allocation score-like framework, such as the use of visualizations like Figure 1 (see [Introduction](#)), may be useful in order to identify “weak points”: locations to which forecasts and policies alike frequently under/overallocate. Such an analysis may be of interest to public health users, as the quality and granularity of forecasts of small-population subgroups (such as counties) has been identified as an area of concern and a potential barrier to forecast adoption. (Rao et al. 2025)

In addition, the distribution schemes scored in this analysis were assumed to come from perfect compliance to forecast predictions and their associated CDFs, which overlooks the complexity of real-world allocation schemes and the nuance with which forecasts are typically interpreted by officials. In practice, public health officials more frequently synthesize multiple information sources and models in order to make resource distribution decisions. [CITE] Furthermore, these decisions may be swayed by political and institutional priorities, emotional reactions, news cycles, or personal bias. [CITE] For this reason, as well as the previously-noted use of abstract resource units in the analysis, we hazard against use of the allocation score to interpret these forecast-based allocations as feasible policies.

Finally, the exclusion of Los Angeles County due to differences in supply lines prevents generalization of the results to forecasting for the state as a whole. Results must be evaluated as pertaining to forecasts made for the entire state minus Los Angeles. **[OPTIONAL: SUPPLEMENTAL MATERIAL WITH SAME ANALYSIS WITH LA COUNTY? COULD RUN BUT NEED TO NOTE VERY BLUNTLY THAT IT SHOULD BE TREATED WITH CAUTION]**

In spite of these limitations, this analysis highlights the successful application of the IAS to an epidemiologic context in which forecasts may be expected to inform allocation-related decisions, as well as possible directions for future work. The frequent failure of the forecasts in this study to outscore a population based-control suggests that it may be a meaningful to add IAS to existing routine forecast evaluation and optimization workflows. **[REQUIRES AN EASIER TO RUN IAS, HOWEVER]** Further, through the use of the COVID-19 Forecasting Hub or a similar platform, would be possible to evaluate and compare multiple forecasts via IAS and produce ensemble forecasts that weight based on recent IAS performance, similarly to how the COVIDhub-trained-ensemble weights its constituent forecasts by the inverse of MWIS. The performance of such IAS-optimized models remains a possible avenue for future forecast development.

Because IAS rewards forecasts that correctly capture relative resource needs across locations, this scoring rule may prove meaningful to public health decision-makers evaluating forecasts as information sources for multi-location resource distribution efforts. This method produces a measurement of avoidable, unaddressed resource need in a scenario where the forecast directly determines policy, rewarding models that capture the relative rankings of needs correctly across the locations of interest. The direct translation of this scoring to politically-relevant metrics, such as money or resource waste, as well as the generation of models that optimize for resource allocation effectiveness, remain areas for future research and development.

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